

Performance Testing

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Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

Performance Testing

Testing approach

The prototype was evaluated through functional and workflow-oriented checks. Because the current implementation calls an external Gemini API and uses local SQLite storage, response time can vary with network/API conditions.

Test ID	Scenario	Expected Result	Result
PT-01	Open home page	Input form loads successfully.	Pass
PT-02	Submit valid user details	7-day workout and nutrition tip are generated and displayed.	Pass
PT-03	Reuse existing User ID	User is directed to the feedback/update flow.	Pass
PT-04	Submit feedback for existing user	Updated workout plan is generated and stored.	Pass
PT-05	Unknown User ID in feedback	Clear error message is displayed.	Pass
PT-06	Admin login with invalid credentials	Login is rejected.	Pass
PT-07	Access admin dashboard without login	User is redirected to admin login.	Pass
PT-08	Admin login and view users	Stored users and plans are displayed.	Pass
PT-09	Admin logout	Session is cleared and user returns to home.	Pass
PT-10	Health endpoint	Returns application status.	Pass

Performance considerations

The main latency contributor is Gemini API generation. SQLite is lightweight for a prototype but may require migration to a production database for larger concurrent workloads. Caching, async API handling, rate limiting and hosted database infrastructure can be considered for scale.